Comm array dish Specification

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Revision	C
Status	APPROVED
Classification	INTERNAL
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1. Purpose

This specification defines the performance, interface and verification requirements for the comm array dish (CM-CMA) of the Communications Relay on Starhaven Orbital Habitat & Shipyard.

2. Scope

Fabrication and installation are performed under work package WP-SHV-CM-03 (WBS 1.13.3), owned by Vance Coldbrook.

The prime supplier for flight hardware is Kuat Precision Instruments. Responsible engineer: Vira Renner.

3. Applicable documents

Document	Title
SHV-SYS-SPEC-0001	Starhaven Master System Specification
SHV-CM-DWG-0202	Comm array dish general arrangement drawing
SHV-CM-BOM-0302	Comm array dish bill of materials
SHV-QA-PLN-0004	Starhaven Verification and Test Plan
SHV-QA-STD-0002	Station Workmanship Standard
SHV-ICD-CN-CM	Interface Control Document: Computer Core & Data Network / Communications Relay

4. Description

Communications covers HoloNet transceivers, comm array dishes and the switches that route passenger, cargo and station traffic.

The comm array dish is delivered as a set of line-replaceable assemblies, integrated on station by Construction Operations and verified by Quality Assurance & Test.

5. Performance requirements

ID	Requirement	Value	Tolerance	Verification
CM-CMA-R01	The comm array dish shall provide a dish diameter of 112 m.	112 m	±5%	Analysis
CM-CMA-R02	The comm array dish shall provide a array count of 12 ea.	12 ea	±2%	Inspection

5.1 General requirements

ID	Requirement	Verification
CM-CMA-G01	Maintainability: Line-replaceable units shall be replaceable by a crew of four within one shift.	Inspection
CM-CMA-G02	Safety: Single faults shall not cause loss of the zone or injury to crew.	Analysis
CM-CMA-G03	Workmanship: Welds and fasteners shall meet the station workmanship standard SHV-QA-STD-0002.	Inspection
CM-CMA-G04	Marking: Every assembly shall carry its drawing number, revision and serial number.	Analysis
CM-CMA-G05	Documentation: Acceptance data packages shall be delivered with every unit.	Inspection

6. Environmental conditions

Condition	Value
Operating pressure	Vacuum to 110 kPa
Radiation, total dose	Station design environment, class R3
Thermal cycling	200 cycles, operating range
Vibration	Station-keeping manoeuvre profile, level 2
Electromagnetic compatibility	Station EMC standard, category B

7. Verification

Each requirement is verified by the method stated in section 5, following SHV-QA-PLN-0004. Acceptance test reports are filed under the CM test-report series.

8. Revision history

Rev	Date	Description	Author
A	2021-10-03	Initial release.	Vira Renner
B	2021-09-29	Clarified verification methods.	Vira Renner
C	2022-05-23	Editorial corrections.	Vira Renner